

Do Heon Han

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Brown University student building LLM retrieval systems, automated reporting pipelines, and 3D scene-generation methods; second author of a paper published at ACM SIGGRAPH Asia 2025.

Education

Brown University

Computer Science and Music (separate degrees) • GPA: 3.87/4.0

Republic of Korea mandatory military service at Yonsei Rehabilitation School - Oct 2023 – Jul 2025; research collaboration with Brown Visual Computing Lab continued remotely throughout.

Providence, RI

Graduation: May 2027

Experience

LG Business Research — AX Group

Full-Time Research Assistant, AI Transformation

LG Group's in-house consulting company.

Seoul, South Korea

Mar 2026 – Aug 2026

- **Proposed and led a project to automate weekly news selection that reflects LG researchers' editorial priorities**, building a LangGraph pipeline with TF-IDF and Korean-language embeddings, exposed as an MCP server for researchers to query directly in Claude.
- **The pipeline surfaced 19% of the topics researchers went on to publish across 11 weeks, 2.8× a prompted-Claude baseline**, with a **22% top-5 hit rate versus the baseline's 11%**.
- Designed source-specific retrieval strategies (regulatory filings, executive changes, CEO statements) and benchmarked them against general-purpose LLM search, which missed roughly 25% of fetched verified events.
- Delivered firm-wide reports and sourced weekly trend publication topics from **roughly 200 conference talks** by building a **multi-stage Python video-to-report pipeline** integrating Parakeet transcription, pyannote diarization, LLM speaker naming, and evidence-grounded summarization.

KB Securities — Global Client Distribution

Full-Time Digital Transformation Intern

South Korea's sixth-largest brokerage by equity

Seoul, South Korea

Aug 2025 – Feb 2026

- Automated morning **news briefings for the Executive Vice President**, distilling ~7,000 **daily RSS entries into ~20 briefing items** by building a Python/LangGraph pipeline with MiniLM deduplication, LLM relevance filtering, and asset-class classification, using free RSS feeds and **Sumy pre-compression** to keep per-run LLM costs negligible.
- Automated daily Excel reports for the **Head of Global Client Distribution**, covering **12,000+ ownership disclosures and treasury-stock filings across 2,600+ companies**, by building a Python ETL pipeline integrating OpenDART, KRX, and Yahoo Finance.
- Reduced weekly report preparation time from **30 minutes to seconds** by building a Python pipeline that automatically organized and condensed Bloomberg economic calendar events into a formatted, readable PowerPoint slide.

Brown Visual Computing Lab

Undergraduate Researcher

- Developed a **3D scene-generation method preferred over state-of-the-art baselines (NVIDIA ATISS, DifuScene) in ~80% of user-study comparisons** by using an LLM to write layout programs supporting arbitrary object types without training on 3D scene datasets.
- Reduced LLM hallucinations to **1.16 per 1,000 generated objects** by combining backtracking, in-context learning, and chain-of-thought prompting.
- Built and iteratively refined an automated LLM evaluator for generated scene quality, improving alignment with human preference judgments by **19 percentage points** from its initial version.

Research Projects & Publications

Procedural Scene Programs for Open-Universe Scene Generation: LLM-Free Error Correction via Program Search

ACM SIGGRAPH Asia 2025 Conference Papers • Second author • <https://github.com/mxgmn/ImperativeScene>

Generates 3D scene layouts with LLM-written procedural programs and corrects layout errors through program search.

Open-Universe Indoor Scene Generation using LLM Program Synthesis and Uncurated Object Databases

arXiv preprint, 2024 • Co-first author

Generates indoor scenes from text using LLM program synthesis, spatial constraint optimization, and retrieval from uncurated 3D object databases.

Skills

Technical: Python, AI Engineering, LangGraph, MCP, LLM evaluation design, Java, C++, Bloomberg Terminal, Excel

Languages: English (native), Korean (native)